

CHIRAG SANJEEVA MURTHY

SOFTWARE DEV ENGINEER II

📞 341-345-1585
✉️ chiragsa93@gmail.com
🏠 New York, NY, USA
🌐 <https://www.linkedin.com/in/chirags-72315089>

PROFESSIONAL OVERVIEW

I specialize in developing scalable cloud solutions and microservices architectures. I've engineered AWS's first Building with Service Teams automation framework, developed Carrier's multi-tenant IoT platform, and architected Sony Pictures' MLOps framework that improved model deployment efficiency by 83% and lead development for TELUS's \$18M Smart Home platform, I hold six AWS certifications and maintain a 9.88/10 CSAT while delivering innovative solutions in cloud architecture, DevOps, and MLOps.

WORK EXPERIENCE

Software Dev Engineer II

Amazon, Inc., New York | 2024 - Present

- Delivered high-quality software features across the full lifecycle, from design and implementation to testing, deployment, and documentation, while adhering to Amazon's coding standards and best practices.
- Coached team members on producing quality code, ensured root cause analysis and permanent fixes for software failures, balanced constraints while avoiding short-term workarounds, wrote clear documentation, sought guidance from senior engineers on tough trade-offs, and trained new team members on software architecture and operations.
- Automated 97% of manual tasks (290/300) in the Cloud Architected Network (CAN) region build process, reducing build time from 8 weeks to 2 weeks.
- Designed and implemented a centralized egress VPC architecture utilizing AWS Network Firewall (ANF), NAT Gateways, and Internet Gateways, supporting up to 1 Tbps of true internet traffic per region.
- Mitigated cross-region S3 and DynamoDB traffic risks by offloading traffic from constrained paths, eliminating regional limits and ensuring service reliability during peak events.
- Led the development of automation tools and optimized pipelines, enabling rapid CAN infrastructure deployment in new regions like NRT and FRA, supporting Amazon's global expansion strategy.
- Spearheaded a GenAI adoption initiative through country-wide workshops, empowering colleagues with cutting-edge skills and driving innovation across the organization.
- Contributed to the "Amazon Region Migration and Flexibility Program" by automating 97% of CAN Core Network SOPs, improving L3 cross-region connectivity automation by 90%.

EDUCATION

BACHELOR OF ENGINEERING, INFORMATION SCIENCE

CMR INSTITUTE OF TECHNOLOGY
2016

PRE-UNIVERSITY DEGREE

CHRIST JUNIOR COLLEGE
2009

SKILLS

- [Cloud] AWS Serverless technologies (Lambda, API Gateway, Cognito, AppSync, SQS, SNS, etc.)
- [Cloud] AWS IoT Services (IoT Core, IoT Rules Engine, IoT Analytics)
- [Cloud] AWS Infrastructure services (EC2, VPC, CloudFormation, Systems Manager Automation, Session Manager, OpsCenter, etc.)
- [Cloud] Workflow services such as Systems Manager Automation, AWS MWAA, AWS Step Functions
- [Cloud] AWS security services (KMS, Secrets Manager, Config, Security Hub)
- [Cloud] Azure IoT/Analytics Services (IoT Hub, PowerBI, Stream Analytics, etc.)
- [Programming languages/frameworks] Python
- [Programming languages/frameworks] Node.js

WORK EXPERIENCE

Senior IoT Architect

AWS Inc., New York | 2024 - 2024

- Accelerated customers' cloud adoption by providing actionable insights, leading multiple projects autonomously.
- Served as technical lead on customer engagements, ensuring architecture quality and guiding technology adoption.
- Demonstrated deep IoT expertise across industries, leveraging business and technology strategies.
- Architected and implemented a comprehensive CI/CD pipeline for a Quality Inspection System (QIS) using AWS services, including CodePipeline, CodeBuild, and CodeCommit, enabling seamless deployment of edge computing and cloud components across multiple environments.
- Designed and deployed a robust security model for factory-to-cloud communication, incorporating AWS IoT Greengrass, custom authentication mechanisms, and end-to-end encryption to ensure data integrity and compliance with customer security requirements.
- Developed and standardized naming conventions and deployment processes for various solution components, including Greengrass devices, cloud backend services, and user interfaces, streamlining development workflows and improving overall system maintainability.

IoT Architect

AWS Inc., New York | 2023- 2024

- Leveraged deep expertise in IoT, software development and DevOps to design and implement solutions that delivered defined business outcomes for customers, collaborating across internal/external teams.
- Contributed insights by analyzing customer needs, recommending suitable technologies/architectures, identifying sales opportunities, developing reusable solutions, creating technical documentation, optimizing processes, participating in recruitment/training, and facilitating alignment in technical discussions.
- Optimized partner consultant onboarding process, reducing engagement manager effort by 93% and interviewer effort by 25%; trained 7-member offshore team to enable independent contributions.
- Spearheaded design and implementation of critical blue/green deployment architecture across 23 microservices, overcoming ambiguity, bridging gaps, and coordinating efforts across 5 workstreams and 20 squads.
- Produced comprehensive documentation, defined standards, established communication protocols adopted program-wide, and provided extended on-site support ensuring timely blue/green migration for employee market trials.

SKILLS

- [languages/frameworks] TypeScript
- [Programming languages/frameworks] Jupyter
- [Programming languages/frameworks] Express
- [Programming languages/frameworks] Plotly
- [CI/CD] CDK/CloudFormation
- [CI/CD] AWS CI/CD suite AWS CodeCommit, CodeBuild, CodeDeploy, CodePipeline
- [CI/CD] Open-source/third party tools - BitBucket, Git, GitHub, GitLab, Apache Airflow
- [Migration] device42, Flexera RISC CloudScape, AWS Application Discovery Service
- [Operating systems] Linux
- [Protocols/standards] GraphQL
- [Protocols/standards] REST
- [Protocols/standards] MQTT
- [Protocols/standards] sparkplugB

WORK EXPERIENCE

Cloud Architect

AWS ProServe India LLP, Bangalore | 2020 - 2023

- Leveraged deep expertise in IoT, software development and DevOps to design and implement secure, scalable, and cost-effective solutions that delivered defined business outcomes for customers, collaborating across internal and external teams.
- Contributed strategic insights by analyzing customer needs, recommending suitable technologies/architectures, identifying sales opportunities, developing reusable solutions, creating technical documentation, optimizing processes, participating in recruitment/training, and facilitating alignment in complex technical discussions.
- Spearheaded pioneering initiatives, securing the first-ever billable AWS Jam engagement and MLOps project for ProServe India, contributing to \$269K in revenue.
- Recognized with multiple awards, including AWS ProServe India MVP of the Year 2020, December 2021 Superstar, and Learning Champion 2022, for outstanding contributions.
- Led a team of consultants in Project Gabba, a \$22M initiative, developing 55 Automation Runbooks with associated Config Rules, reducing customer implementation time by 240% and serving 455 unique AWS customers in a single month.
- Designed and implemented a multi-tenant authentication and authorization strategy for Carrier's connected devices platform, collaborating with the IoT Core Identity service team to shape a new feature implementation.
- Architected a foundational MLOps framework for Sony Pictures Network India, improving model production time by 97%, reducing monitoring costs by 70%, and enhancing overall operational efficiency by 83%.
- Developed a multi-Control Tower Customizations framework for Genpact's DC migration, enabling the deployment of over 100 security controls across multiple AWS environments, facilitating the migration of 30+ workloads to AWS.

AWARDS & ACKNOWLEDGEMENTS

- AWS ProServe India MVP of the Year 2020
- AWS Certified Solutions Architect Professional
- AWS Certified Security Specialty
- AWS Certified Solutions Architect Associate
- AWS Certified Developer Associate
- AWS Certified SysOps Administrator – Associate
- AWS Certified Cloud Practitioner
- AWS ProServe India December 2021 Superstar
- MCPS: Microsoft Certified Professional - G152-4317
- MS: Design and Implement Big Data Analytics Solutions - G152-4319

WORK EXPERIENCE

Cloud Consultant

AWS ProServe India LLP, Bangalore | Jul 2019 – Jul 2020

- Primary responsibilities include assisting in delivery, pre-sales as well as IP creation in the form of reusable assets and documents. Also involved in training customer, partner and team mates on AWS best practices and services.
- Multi-Control Tower Baseline Automation: Genpact wants all-in migration of 3,300 servers across 10 data centers to AWS by 2022. Increase automation of operational and DevOp processes. Responsible for Multi-Control Tower Customization Framework Solution for large customers. Built cfn-guard rules spanning 25 services to improve security.
- Altisource migration: Responsible for identifying, collecting and organizing customer's on-premise infrastructure spanning two datacenters (approx. 700 servers) during Migration Readiness and Planning phase.
- Tata Sky Security Review: Conducted security assessment for customer's data warehouse migration from Netezza to Amazon Redshift. Produced best practice recommendations and prioritized security vulnerabilities.
- AWS Config 50/500 project: Led SSM automation document development efforts. Designed, developed, reviewed, and deployed public AWS artifacts. Produced over 55 automation runbooks as response actions for non-compliant AWS config rules or Security Hub controls. Artifacts engaged over 11,518 times across 455 accounts in March 2021.
- MongoDB and Elasticsearch migration: Responsible for migration pattern design and implementation for MongoDB and Elasticsearch workloads. Customer requirements included zero downtime, one-click rollback, on-premise cluster to remain active and primary, no managed services, and EC2 instances to serve queries only during testing periods.

Cloud Support Associate II

Amazon Web Services India Private Limited, Bangalore | Dec 2018 - Jul 2018

- Primary responsibilities include assisting customers with solving and troubleshooting complex issues on the AWS platform as well as build knowledge base for other engineers through the release of KNET courses and articles. Focus was on Developer and Mobile Services.
- Provided guidance and resolution for more complex customer cases for other engineers through SupportOps programme – a support engineer team with elevated privileges. Was a part of four SupportOps teams – SNS, IoT, SQS, Pinpoint.

WORK EXPERIENCE

Cloud Support Associate I

Amazon Web Services India Private Limited, Bangalore | Jan 2018 - Nov 2018

- Primary responsibilities include assisting customers with troubleshooting technical issues and working with other engineers and service teams to reproduce and resolve issues or deploy changes to AWS services.
- Trained new and tenured engineers on IoT concepts and AWS IoT services to cope with the growing need for IoT experts in AWS Premium Support.
- Was a part of the initial team for RCS Flex (ProServe programme) in Bangalore. Designed and developed three AWS Config rules for external customers on behalf of the ProServe team.
- Actively contributed to Forum Support where non-Premium Support customers post queries/concerns/doubts.

Solutions Engineer - Cloud & IoT

CloudThat Technologies, Bangalore | Sep 2016 - Jan 2018

- Primary responsibilities include assisting in delivery, pre-sales as well as IP creation in the form of reusable assets and documents. Also involved in training customer, partner and team mates on AWS best practices and services.
- Developed the backend for a modular telepresence robot. The robot was mainly geared toward telepresence. The robot was also capable of rudimentary cloud deployments such as VM and Storage creation to aid Cloud Architects. The backend handled the cognitive aspect (Natural Language Processing, Speech to Text, Text to Speech, Face Detection & Recognition and Voice Recognition) as well as the information processing (database operations).
- Technologies implemented included Azure, SQL, Lighttpd and several others. Programming languages deployed were Python and Node.js.

Intern - Cloud & IoT

CloudThat Technologies, Bangalore | Jul 2016 - Sep 2018

- Developed curriculums for IoT (corporate and retail) with varying degrees of focus starting from the abstract level to end-to-end solution development.
- Developed Azure curriculum for corporate and retail batches with prime focus on the architecture aspect of Cloud deployments.